

3D reasoning. It uses a depth predictor and CNN to extract spatial local features and uses learnable 3D query tokens for geometric coordinate regression. It combines black-box networks and white-box projection to address changes in camera focal lengths.

3.1.3 RGB-D Image as Input

Depth is estimated in SpatialPIN [Ma et al. \[2024a\]](#) by ZoeDepth when finding field of view (FOV) through perspective fields and provided for 3D-scene understanding and reconstruction. M3D-LaMed [Bai et al. \[2024\]](#) pre-trains the 3D medical vision encoder with medical image slices along depth and introduces end-to-end tuning to integrate 3D information into LLM.

3.1.4 3D Medical Image as input

Unlike previous research focused on 2D medical images, integrating multi-modal other information such as textual descriptions, M3D-LaMed [Bai et al. \[2024\]](#) is specifically designed for 3D CT images by analyzing spatial features. It demonstrates excellent performance across multiple tasks, including image-text retrieval, report generation, visual question answering, localization, and segmentation. In order to generate radiology reports automatically, a brand-new framework [Liu et al. \[2024a\]](#) is proposed to employ low-resolution (LR) visual tokens as queries to extract information from high-resolution (HR) tokens, ensuring that detailed information is retained across HR volumes while minimizing computational costs by processing only the HR-informed LR visual queries. 3D-CT-GPT [Chen et al. \[2024b\]](#), based medical visual language model, is tailored for the generation of radiology reports from 3D CT scans, with a focus on chest CTs. OpenMEDLab [Wang et al. \[2024\]](#) comprises and publishes a variety of medical foundation models to process multi-modal medical data including Color Fundus Photography (CFP), Optical Coherence Tomography (OCT), endoscopy videos, CT&MR volumes and other pathology images.

3.1.5 Discussion

Image-based spatial reasoning methods offer significant advantages, such as easy data acquisition and integration with pre-trained 2D models. Multi-view images provide rich spatial information, while depth estimation enhances scene understanding. However, challenges remain, including limited depth from single views, scale uncertainty, occlusion, and viewpoint dependency. These methods also face issues with visual hallucinations, generalization to novel scenes, and high computational costs. Future research should focus on improving multi-view integration and depth estimation to address these limitations.

3.2 Recent Advances of Point Cloud-based Spatial Reasoning

As shown in Figure 4, point cloud-based spatial reasoning has advanced significantly in recent years, employing three main alignment methods: Direct, Step-by-step, and Task-specific Alignment. These methods are essential for integrating point cloud data with language models to enable effective spatial reasoning. Direct Alignment establishes immediate connections between point cloud features and language model em-

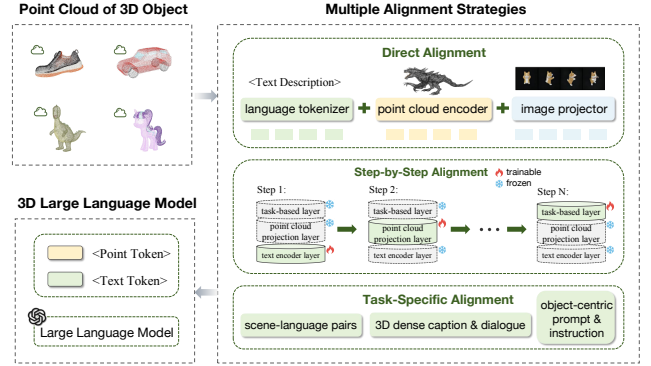


Figure 4: An overview of point cloud-based approaches.

beddings, while Step-by-step Alignment follows a sequential process through multiple stages. Task-specific Alignment is customized for particular spatial reasoning requirements. The choice of method depends on specific application needs and constraints.

3.2.1 Direct Alignment

Direct alignment methods create direct connections between point cloud data and language models. PointCLIP [\[Zhang et al., 2022\]](#) was a pioneer, projecting point clouds into multi-view depth maps and using CLIP’s pre-trained visual encoder for feature extraction, which was then aligned with textual features through a hand-crafted template. This approach showed promising results in zero-shot and few-shot classification tasks by transferring 2D knowledge to the 3D domain. PointCLIP V2 [\[Zhu et al., 2023\]](#) improved the projection quality with a realistic projection module and used GPT-3 for generating 3D-specific text descriptions, achieving better performance in zero-shot classification, part segmentation, and object detection. Chat-Scene [\[Huang et al., 2024\]](#) introduced object identifiers to facilitate object referencing during user-assistant interactions, representing scenes through object-centric embeddings. PointLLM [\[Xu et al., 2025\]](#) advanced the field by integrating a point cloud encoder with a powerful LLM, effectively fusing geometric, appearance, and linguistic information, and overcoming data scarcity with automated generation. These methods demonstrate the potential for effective 3D point cloud understanding through language models, enabling improved spatial reasoning and human-AI interaction.

3.2.2 Step-by-step Alignment

Step-by-step alignment has gained popularity in integrating point cloud features with language models. Notable approaches include GPT4Point [\[Qi et al., 2024b\]](#), which uses a Bert-based Point-QFormer for point-text feature alignment, followed by object generation. Grounded 3D-LLMs [\[Chen et al., 2024d\]](#) first aligns 3D scene embeddings with textual descriptions via contrastive pre-training, then fine-tunes with referent tokens. LiDAR-LLMs [\[Yang et al., 2023\]](#) employ a three-stage process: cross-modal alignment, object-centric learning, and high-level instruction fine-tuning. MiniGPT-3D [\[Tang et al., 2024a\]](#) follows a four-stage strategy, from point cloud projection to advanced model enhancements using Mixture of Query Experts. GreenPLM [\[Tang et al., 2024b\]](#) uses

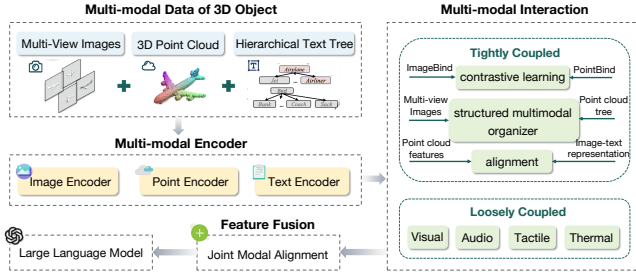


Figure 5: An overview of hybrid modality-based approaches.

a three-stage method that aligns a text encoder with an LLM using large text data, followed by point-LLM alignment with 3D data. These step-by-step approaches highlight the gradual improvement of spatial reasoning in 3D contexts, offering valuable insights for future research.

3.2.3 Task-specific Alignment

Task-specific alignment customizes models for specific spatial reasoning tasks to improve performance and generalization. SceneVerse [Jia *et al.*, 2024] introduces a large 3D vision-language dataset and Grounded Pre-training for Scenes (GPS), using multi-level contrastive alignment for unified scene-text alignment, achieving state-of-the-art results in tasks like 3D visual grounding and question answering. LL3DA [Chen *et al.*, 2024c] presents a dialogue system that integrates textual instructions and visual interactions, excelling in complex 3D environments. Chat-3D [Wang *et al.*, 2023] proposes a three-stage training scheme to align 3D scene representations with language models, capturing spatial relations with limited data. VisProg [Yuan *et al.*, 2024] introduces visual programming for zero-shot open-vocabulary 3D grounding, leveraging LLMs to generate and execute programmatic representations. These task-specific approaches highlight the importance of adapting models to complex spatial relationships, enabling robust performance even with limited data or zero-shot tasks.

3.2.4 Discussion

The three alignment approaches—Direct, Step-by-step, and Task-specific—each have distinct strengths and challenges. Direct alignment offers efficiency and quick results but struggles with complex spatial relationships. Step-by-step alignment improves feature integration at the cost of higher computational resources and training time. Task-specific alignment excels in specialized tasks but may lack broader applicability.

3.3 Hybrid Modality-based Spatial Reasoning

Hybrid modality-based spatial reasoning integrates point clouds, images, and LLMs through Tightly Coupled and Loosely Coupled approaches, as shown in Figure 5. The Tightly Coupled approach fosters close integration, enabling seamless interaction and high performance, while the Loosely Coupled approach promotes modularity, allowing independent operation of components for greater scalability and flexibility at the cost of reduced real-time interaction.

3.3.1 Tightly Coupled

Several recent works have explored tightly integrated approaches for spatial reasoning across point clouds, images and

language modalities: Point-Bind [Guo *et al.*, 2023] proposes a joint embedding space to align point clouds with images and text through contrastive learning. It leverages ImageBind to construct unified representations that enable tasks like zero-shot classification, open-world understanding and multi-modal generation. The tight coupling allows Point-Bind to reason about point clouds using both visual and linguistic cues. JM3D [Ji *et al.*, 2024] introduces a Structured Multimodal Organizer that tightly fuses multi-view images and hierarchical text trees with point clouds. This coupled architecture enables detailed spatial understanding by leveraging complementary information across modalities. The Joint Multi-modal Alignment further enhances the synergistic relationships between visual and linguistic features. Uni3D [Zhou *et al.*, 2023] employs a unified transformer architecture that directly aligns point cloud features with image-text representations. By tightly coupling the modalities through end-to-end training, it achieves strong performance on tasks like zero-shot classification and open-world understanding. The shared backbone enables efficient scaling to billion-parameter models. Uni3D-LLM [Liu *et al.*, 2024b] extends this tight coupling to LLMs through an LLM-to-Generator mapping block. This enables unified perception, generation and editing of point clouds guided by natural language. The tight integration allows leveraging rich semantic knowledge from LLMs while maintaining high-quality 3D understanding.

3.3.2 Loosely Coupled

Loosely coupled approaches maintain greater independence between different modalities while still enabling interaction through well-defined interfaces. MultiPLY [Hong *et al.*, 2024] proposes a multisensory embodied LLM that handles multiple input modalities (visual, audio, tactile, thermal) through separate encoders. The modalities are processed independently and communicate through action tokens and state tokens. This decoupled design allows the system to process each modality with specialized encoders optimized for that data type, while enabling scalability and modularity in the system architecture. Similarly, UniPoint-LLM [Liu *et al.*] introduces a Multimodal Universal Token Space (MUTS) that loosely connects point clouds and images through independent encoders and a shared mapping layer. This modular design allows easy integration of new modalities and simplified training by only requiring alignment between new modalities and text, rather than pairwise alignment between all modalities. The main benefits of loosely coupled architectures include greater modularity and flexibility in system design, easier integration of new modalities, and independent scaling of different components. However, this approach may result in less optimal joint representation learning, reduced real-time interaction capabilities, and potential information loss between modalities compared to tightly coupled approaches.

3.3.3 Discussion

The choice between tightly and loosely coupled approaches presents important tradeoffs in multimodal spatial reasoning systems. Tightly coupled approaches like Point-Bind and JM3D offer stronger joint representation learning and real-time interaction capabilities through end-to-end training and shared feature spaces. This makes them particularly suitable

	Model	Data Source	Alignment Type	Pre-training	Fine-tuning	Task	Code
Image - based	LLaVA-3D [Zhu <i>et al.</i> , 2024b]	Multi-view Images	-	✓	✓	3D VQA, 3D Scene Understanding	code
	Agent3D-Zero [Zhang <i>et al.</i> , 2024]	Multi-view Images	-	✓	✗	3D VQA, 3D Semantic Segmentation	✗
	ShapeLLM [Qi <i>et al.</i> , 2024a]	Multi-view Images	-	✓	✓	3D Object Classification, 3D Scene Captioning	code
	Scene-LLM [Fu <i>et al.</i> , 2024]	Multi-view Images	-	✓	✓	3D VQA, Dense Captioning	✗
	SpatialPIN [Ma <i>et al.</i> , 2024a]	RGB-D Images	-	✓	✗	3D Motion Planning, Task Video Generation	✗
	LLM3D [Yang <i>et al.</i> , 2024]	Monocular Images	-	✓	✓	3D Grounding, 3D VQA	✗
	Spatialvlm [Chen <i>et al.</i> , 2024a]	Monocular Images	-	✓	✓	Dense Reward Annotator, Spatial Data Generation	code
	M3D-LaMed [Bai <i>et al.</i> , 2024]	Medical Images	-	✓	✓	3D VQA, 3D VLP	code
	HILT [Liu <i>et al.</i> , 2024a]	Medical Images	-	✓	✓	3DHRG	✗
	3D-CT-GPT [Chen <i>et al.</i> , 2024b]	Medical Images	-	✓	✓	Radiology Report Generation, 3D VQA	✗
	OpenMEDLab [Wang <i>et al.</i> , 2024]	Medical Images	-	✓	✓	Medical Imaging	code
Point Cloud - based	PointLLM [Xu <i>et al.</i> , 2025]	Point Cloud	Direct Alignment	✓	✓	3D Object Classification, 3D Object Captioning	code
	Chat-Scene [Huang <i>et al.</i> , 2024]	Point Cloud	Direct Alignment	✓	✓	3D Visual Grounding, 3D Scene Captioning	code
	PointCLIP [Zhang <i>et al.</i> , 2022]	Point Cloud	Direct Alignment	✓	✓	3D Point Cloud Classification	code
	PointCLIPv2 [Zhu <i>et al.</i> , 2023]	Point Cloud	Direct Alignment	✓	✓	3D Point Cloud Classification	code
	GPT4Point [Qi <i>et al.</i> , 2024b]	Point Cloud	Step-by-step Alignment	✓	✓	3D Object Understanding	code
	MiniGPT-3D [Tang <i>et al.</i> , 2024a]	Point Cloud	Step-by-step Alignment	✓	✓	3D Object Classification, 3D Object Captioning	code
	GreenPLM [Tang <i>et al.</i> , 2024b]	Point Cloud	Step-by-step Alignment	✓	✓	3D Object Classification	code
	Grounded 3D-LLM [Chen <i>et al.</i> , 2024d]	Point Cloud	Step-by-step Alignment	✓	✓	3D Object Detection, 3D VQA	code
	Lidar-LLM [Yang <i>et al.</i> , 2023]	Point Cloud	Step-by-step Alignment	✓	✓	3D Captioning, 3D Grounding	code
	3D-LLaVA [Deng <i>et al.</i> , 2025]	Point Cloud	Task-specific Alignment	✓	✓	3D VQA, 3D Captioning	code
	ScanReason [Zhu <i>et al.</i> , 2024a]	Point Cloud	Task-specific Alignment	✓	✓	3D Reasoning Grounding	code
	SegPoint [He <i>et al.</i> , 2024]	Point Cloud	Task-specific Alignment	✓	✓	3D Instruction Segmentation	✗
	Kestrel [Fei <i>et al.</i> , 2024]	Point Cloud	Task-specific Alignment	✓	✓	Part-Aware Point Grounding	✗
	SIG3D [Man <i>et al.</i> , 2024]	Point Cloud	Task-specific Alignment	✓	✓	Situation Estimation	code
	Chat-3D [Wang <i>et al.</i> , 2023]	Point Cloud	Task-specific Alignment	✓	✓	3D VQA	code
	LL3DA [Chen <i>et al.</i> , 2024c]	Point Cloud	Task-specific Alignment	✓	✓	3D Dense Captioning	code
Hybrid - based	Point-bind [Guo <i>et al.</i> , 2023]	Point cloud, Image	Tightly Coupled	✓	✓	3D Cross-modal Retrieval, Any-to-3D Generation	code
	JM3D [Ji <i>et al.</i> , 2024]	Point cloud, Image	Tightly Coupled	✓	✓	Image-3D Retrieval, 3D Part Segmentation	code
	Uni3D [Zhou <i>et al.</i> , 2023]	Point cloud, Image	Tightly Coupled	✓	✓	Zero-shot Shape Classification	code
	Uni3D-LLM [Liu <i>et al.</i> , 2024b]	Point cloud, Image	Tightly Coupled	✓	✓	3D VQA	✗
	MultiPLY [Hong <i>et al.</i> , 2024]	Point cloud, Image	Loosely Coupled	✓	✓	Object retrieval	code
	UniPoint-LLM [Liu <i>et al.</i>]	Point cloud, Image	Loosely Coupled	✓	✓	3D generation, 3D VQA	✗

Table 1: Taxonomy of Large Language Models with spatial reasoning capability. This table presents a comprehensive comparison of various 3D vision-language models categorized by their input modalities (image-based, point cloud-based, and hybrid-based), showing their data sources, alignment types, training strategies (pre-training and fine-tuning), primary tasks, and code availability. The models are organized into three main categories based on their input type: image-based models, point cloud-based models, and hybrid models that utilize both modalities.

for applications requiring detailed spatial understanding and precise control. However, they can be more complex to train and scale, and adding new modalities may require significant architectural changes. In contrast, loosely coupled approaches like MultiPLY and UniPoint-LLM provide greater modularity and flexibility, making them easier to extend and maintain. They allow independent optimization of different components and simplified training procedures, but may sacrifice some performance in tasks requiring fine-grained cross-modal understanding. The optimal choice ultimately depends on specific application requirements - tightly coupled architectures may be preferred for specialized high-performance systems, while loosely coupled designs better suit general-purpose platforms prioritizing extensibility and maintainability. Future work may explore hybrid approaches that combine the benefits of both paradigms, potentially using adaptive coupling mechanisms that adjust based on task demands.

4 Applications

A key research focus leverages LLMs to enhance robotic embodied intelligence, enabling machines to interpret natural language commands for real-world tasks. This includes robotic control, navigation, and manipulation, where LLMs parse instructions, generate action plans, and adapt to dynamic environments—for instance, guiding robots to locate objects in cluttered spaces using text-based prompts.

3D Scene Understanding. Advanced 3D scene analysis integrates multimodal data (e.g., images, point clouds, text) for

tasks like open-vocabulary segmentation, semantic mapping, and spatial reasoning. Central to this is 3D visual question answering (3D-VQA), requiring models to interpret queries about object attributes, spatial relationships, or contextual roles within scenes. Context-aware systems further account for user perspectives to deliver precise responses.

Cross-Domain Applications. In healthcare, LLMs analyze volumetric medical scans (e.g., CT) for lesion detection and automated diagnostics. Autonomous driving systems utilize 3D-capable LLMs to interpret traffic scenes, aiding object detection [Zha *et al.*, 2023, 2024] and path planning. Design-oriented applications include generating indoor layouts from textual requirements, while educational tools employ interactive 3D environments to teach spatial concepts.

5 Challenges and Future Directions

Table 1 summarizes the models that leverage LLMs to assist graph-related tasks according to the proposed taxonomy. Based on the above review and analysis, we believe that there is still much space for further enhancement in this field. Recent advances in integrating LLMs with three-dimensional (3D) data have demonstrated considerable promise. However, numerous challenges must still be overcome to realize robust and practical 3D-aware LLMs. Below, we summarize these obstacles and then outline potential pathways to address them, highlighting key research directions for the future.